

Gate Class Documentation

Source File: Split.h
Namespace: cap
Class Header: class Split : public Dev

Overview

The *Split* class is *Dev* object container for another *Dev* object that can be used as an input for at most 20 other *Dev* objects.

Constructors

- Split() (default constructor)
 - **Purpose:** Creates an empty split.
- Split(const Split& obj) (copy constructor) [deleted]

Destructor

- ~Split() [virtual]
 - **Purpose:** It unbinds input.

Assignment Operator

- operator=(const Split& rhs) [deleted]

Member Functions

- bind(Dev& obj) [private final]
 - **Purpose:** Bind a parent function to it and returns true only if it has more space and *obj* is not already a parent; otherwise, it returns false.
 - **Parameter(s):**
 - *obj*: *Dev* reference object.
 - **Return:** Boolean.
- unbind() [private final]
 - **Purpose:** Unbind all parent functions.
- broken() const [final]
 - **Purpose:** Returns true if it is broken or any of its parents are broken; otherwise, it returns false.
 - **Return:** Boolean.
- reset() [final]
 - **Purpose:** Removes its input and parent functions.
- input(Dev& obj)
 - **Purpose:** Makes *obj* its input and returns true only if it is not broken and *obj* is not bounded or bounded to it; otherwise, it returns false.
 - **Parameter(s):**
 - *obj*: *Dev* reference object.
- count() const
 - **Purpose:** Returns the number of parent functions.
 - **Return:** Unsigned integer.
- bounded() const [final]
 - **Purpose:** Returns true only if it cannot bind to any more parent functions.
 - **Return:** Boolean.
- valid() const [final]
 - **Purpose:** Returns true only if it is not broken and has an input.
 - **Return:** Boolean.

- `output()` `const` `[final]`
 - **Purpose:** Returns the output of its input if it is valid; otherwise, returns false.
 - **Return:** Boolean.
- `toString()` `const` `[final]`
 - **Purpose:** Returns string representation of its input if it is valid; otherwise, returns "X".
 - **Return:** String.